

WHAT IS PICOTOOL?

Picotool is a **command-line utility** provided by the Raspberry Pi Foundation to work with Raspberry Pi Pico and other **RP2040 boards**.

WHAT IS THE USE OF PICOTOOL?

- Helps in **debugging** and **maintenance** by showing metadata and board info.
- Supports advanced features like **encryption**, **signing**, and **config storage** for professional projects.
- **cp** simply copies a UF2/BIN file to the Pico's USB mass storage, while **picotool** can flash, read, and manage firmware with extra features like metadata, encryption, and info display.

DIFFERENCE BETWEEN CP AND PICOTOOL COMMAND

cp	picotool
Just copies the .uf2 file to the Pico when it shows up as a USB drive (RPI-RP2) in BOOTSEL mode .	Talks directly to the Pico's bootloader over USB and programs flash memory (no need for it to appear as a drive).
Works only with .uf2 files .	Works with .uf2 and raw .bin files (more flexible).
No verification — it just copies the file.	Can verify, inspect program info, dump memory,

BASIC COMMANDS IN PICOTOOL

picotool help info

picotool info displays details about the connected Pico board and its firmware, including metadata like program name, version, and build date.

-b, --basic

Display basic information

sudo picotool info -b

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$ sudo picotool info -b
Program Information
name:      blink
web site:  https://github.com/raspberrypi/pico-examples/tree/HEAD/blink
binary start: 0x10000000
binary end:   0x10001b28
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$
```

-d, --device

Include device information

sudo picotool info -d

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$ sudo picotool info -d
Device Information
type:      RP2040
flash size: 2048K
flash id:   0xE661640843299639
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$
```

-l, --build

Include build attributes

```
sudo picotool info -l
```

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$ sudo picotool info -l
Build Information
sdk version:      2.2.0
pico_board:      pico
boot2_name:      boot2_w25q080
build date:      Oct  1 2025
build attributes: Release
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$
```

-a, --all

Include all available information about the connected Pico, including embedded metadata like program name, version, and build date.

```
sudo picotool info -a
```

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$ sudo picotool info -a
Program Information
name:          blink
web site:      https://github.com/raspberrypi/pico-examples/tree/HEAD/blink
binary start:  0x10000000
binary end:    0x10001b28

Fixed Pin Information
none

Build Information
sdk version:    2.2.0
pico_board:    pico
boot2_name:     boot2_w25q080
build date:     Oct  1 2025
build attributes: Release

Device Information
type:           RP2040
revision:       B2
flash size:     2048K
flash id:       0xE661640843299639

Metadata Blocks
none
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build$
```

picotool config

picotool config is used to read or write custom metadata (key-value pairs) into a Pico's firmware or directly onto the board.

[Key-value pairs are a way to store data as a label (key) and its corresponding value.]

sudo picotool config blink.uf2

`sudo picotool config blink.uf2` displays all key-value metadata embedded inside the blink.uf2 firmware file.

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$ sudo picotool config blink.uf2
File blink.uf2:
```

picotool load

picotool load flashes a firmware file (UF2, BIN, or ELF) onto the Raspberry Pi Pico's flash memory.

sudo picotool load blink.uf2 -f

sudo picotool load blink.uf2 -f flashes the blink.uf2 firmware onto the connected Pico, forcing overwrite if needed.

picotool save

picotool save copies the contents of the Pico's flash memory to a file (UF2 or BIN) for backup or inspection.

sudo picotool save -a blink.uf2

Saves a complete copy of everything in the Pico's flash memory into the file blink.uf2. This command overwrites the original compiled program with the current flash contents.

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$ sudo picotool save -a blink.uf2
Saving file: [=====] 100%
Wrote 4194304 bytes to blink.uf2
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$
```

picotool verify

picotool verify is to check that the firmware file on your computer exactly matches what is flashed on the Pico, ensuring it was written correctly.

sudo picotool verify blink.uf2

Checks that the firmware (blink.uf2) matches what is currently stored on the Pico's flash memory.

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$ sudo picotool verify blink.uf2
Verifying Flash:      [=====] 100%
OK
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$
```

picotool reboot

picotool reboot is used to restart the connected Pico, either into the application program or into USB bootloader mode.

sudo picotool reboot -a

sudo picotool reboot -a restarts the Pico so it boots directly into and runs the flashed application program.

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$ sudo picotool reboot -a
^The device was rebooted into application mode.
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$
```

sudo picotool reboot -u

sudo picotool reboot -u restarts the Pico and forces it into USB bootloader mode, making it appear as a USB mass storage device so you can safely upload or update firmware.

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$ sudo picotool reboot -u
The device was rebooted into BOOTSEL mode.
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$
```

picotool version

sudo picotool version

sudo picotool version displays the installed Picotool utility's version information.

```
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$ sudo picotool version
picotool v2.2.0-a4 (Linux, GNU-13.3.0, Release)
dubi@dubi-Acer-ALG-AL15G-52:~/pico/pico-examples/build/blink$
```

COMMANDS AND ITS USES:

1. **info** → Shows information from the connected device or a file. Without arguments, it lists basic info for all RP2040 devices in BOOTSEL mode.
2. **config** → View or modify program configuration settings on the device or firmware file.
3. **load** → Flash a program or memory range from a file onto the Pico.
4. **encrypt** → Encrypt a program file for secure deployment.
5. **seal** → Add final metadata to a binary, optionally including a hash and/or signature.
6. **link** → Combine multiple binaries into a single memory block.

7. **save** → Backup the program or flash memory from the device to a file.
8. **verify** → Confirm that the device's memory matches the contents of a file.
9. **reboot** → Restart the Pico.
10. **otp** → Manage RP2040 One-Time-Programmable (OTP) memory.
11. **partition** → Work with RP2040 partition tables.
12. **uf2** → Commands for creating or checking UF2 files.
13. **version** → Show the installed Picotool version.
14. **coprodis** → Post-process coprocessor instructions in disassembly files.
15. **help** → Display general help or help for a specific command.